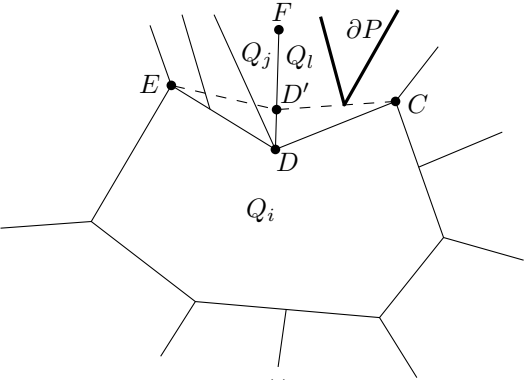
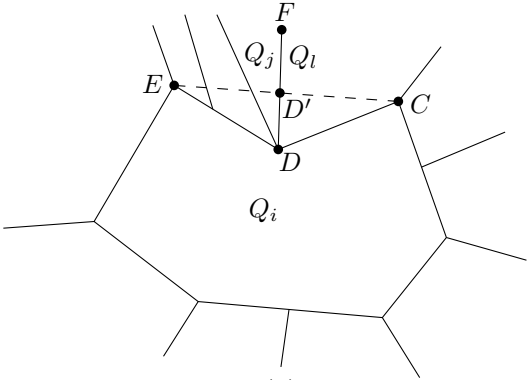
(i)

(ii)

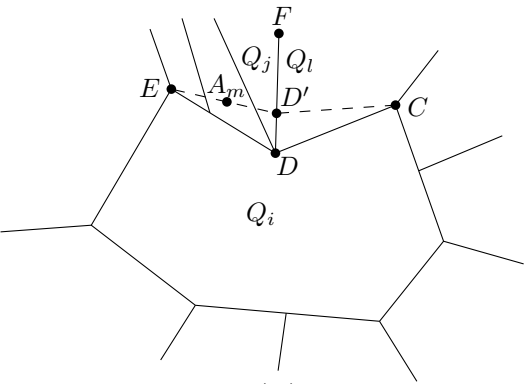
(iii)


F
Q_j
Q_l
∂P
E
D'
C
D
Q_i

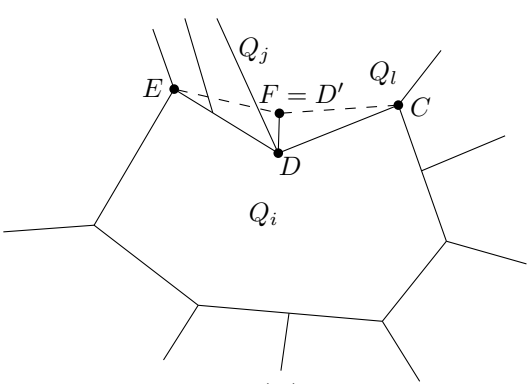
(i)


F
Q_j
Q_l
E
D'
C
D
Q_i

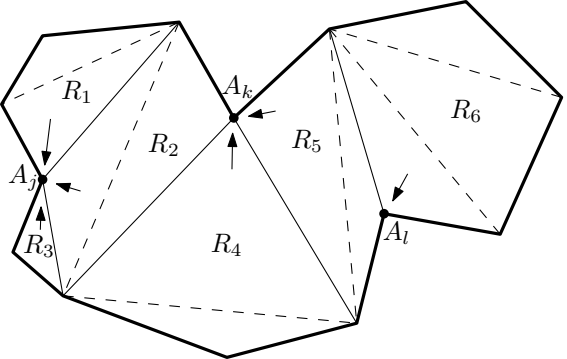
(ii)

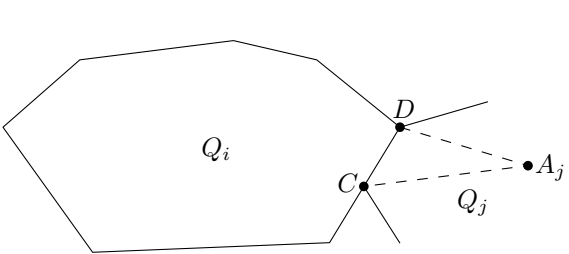

F
Q_j
Q_l
E
A_m
D'
C
D
Q_i

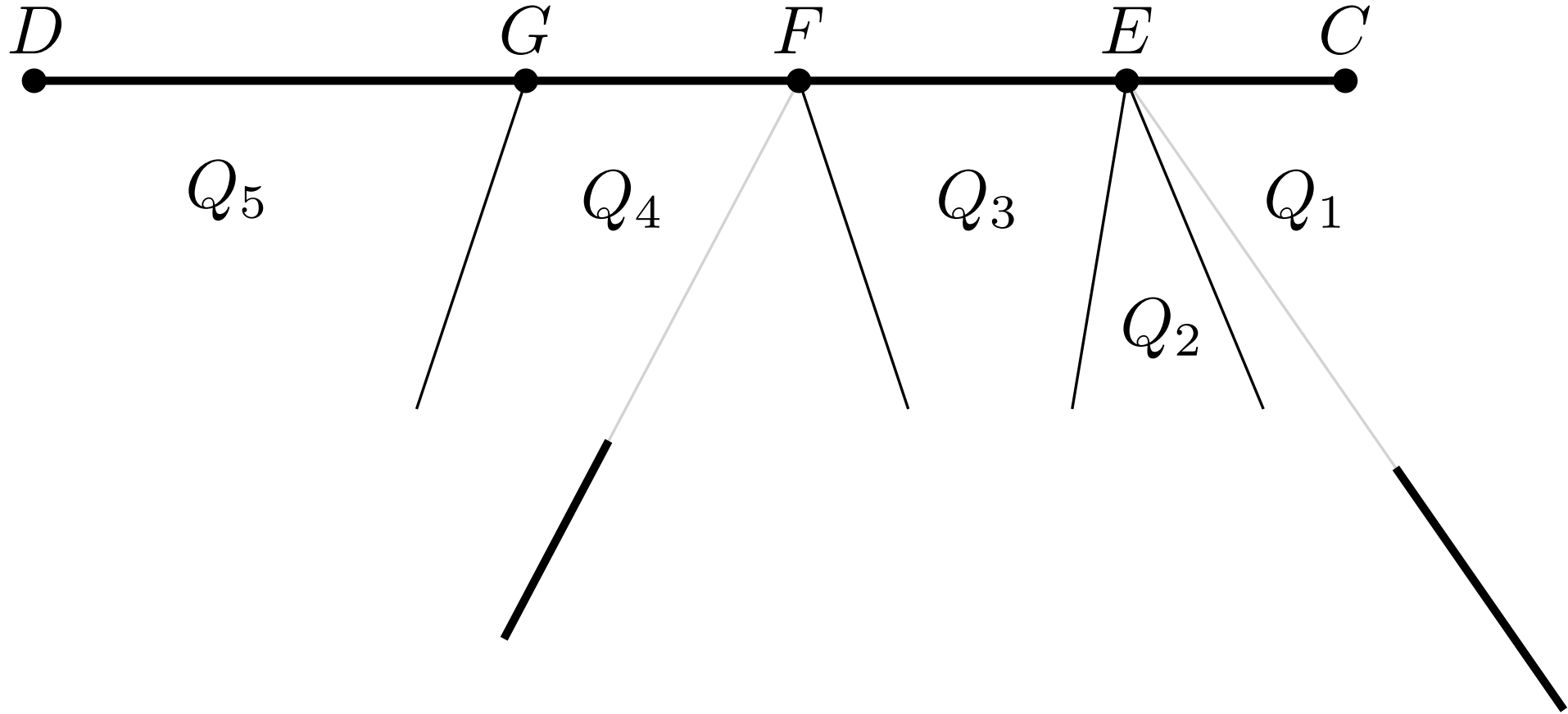
(iii)

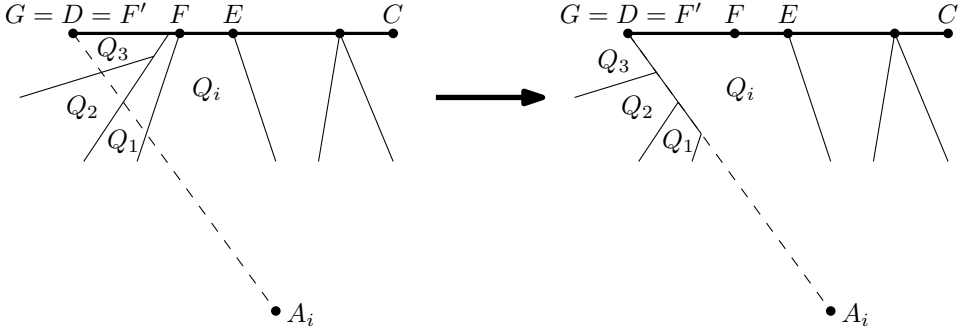

Q_j
Q_l
E
$F = D'$
C
D
Q_i

(iv)

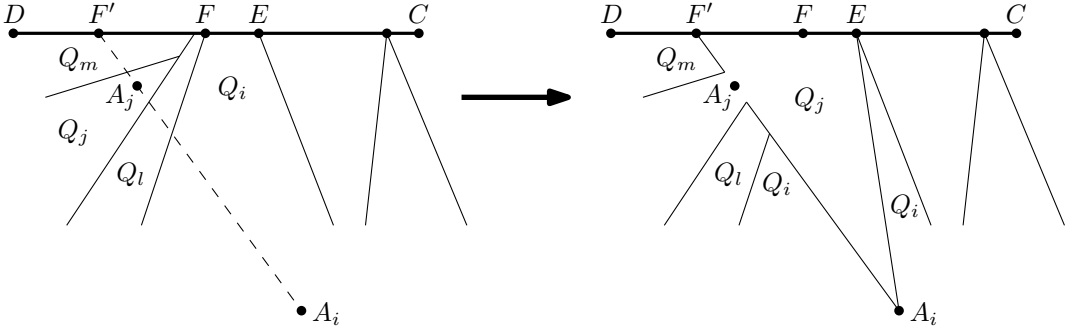

R_1
R_2
R_3
R_4
R_5
R_6
A_j
A_k
A_l

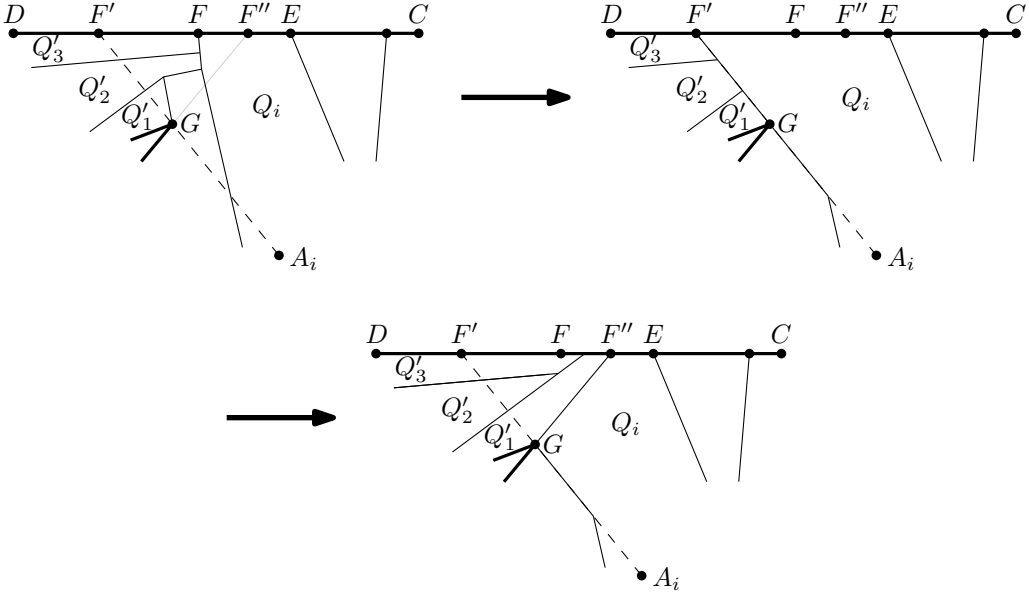

Q_i
Q_j
D
C
A_j

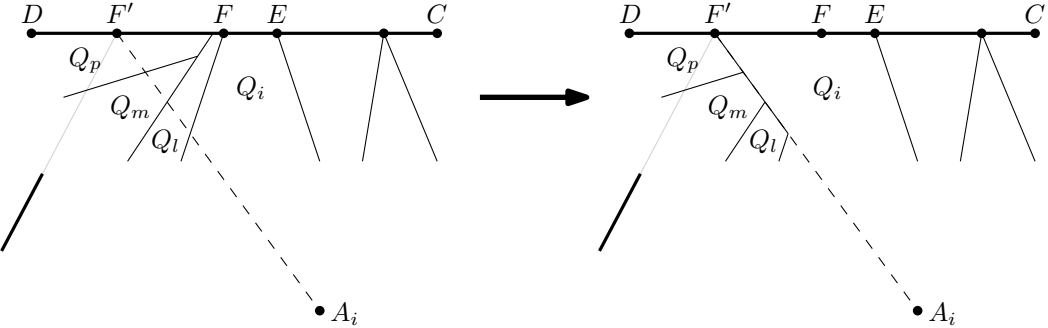

D
G
F
E
C
Q_5
Q_4
Q_3
Q_2
Q_1

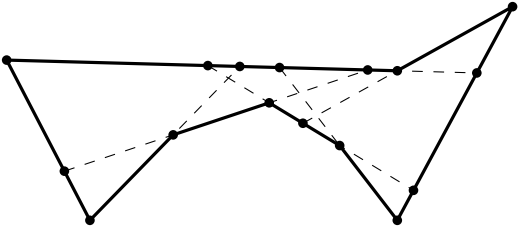


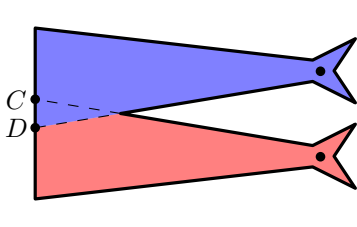
Obsolete, should be covered by case (i).

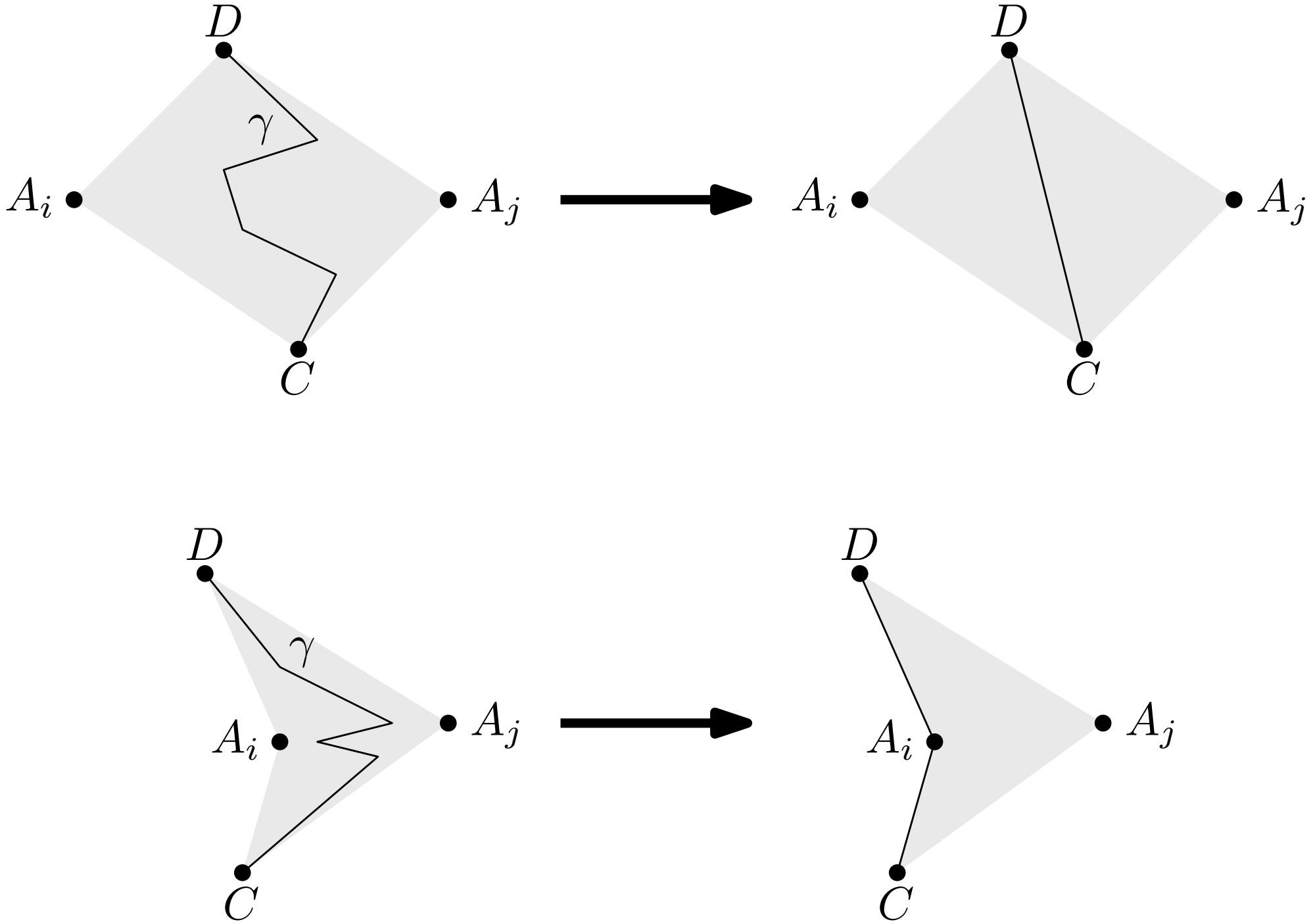

D
F′
F
E
C
Q_m
A_j
Q_i
Q_j
Q_l
A_i
D
F′
F
E
C
Q_m
A_j
Q_j
Q_l
Q_i
Q_i
A_i

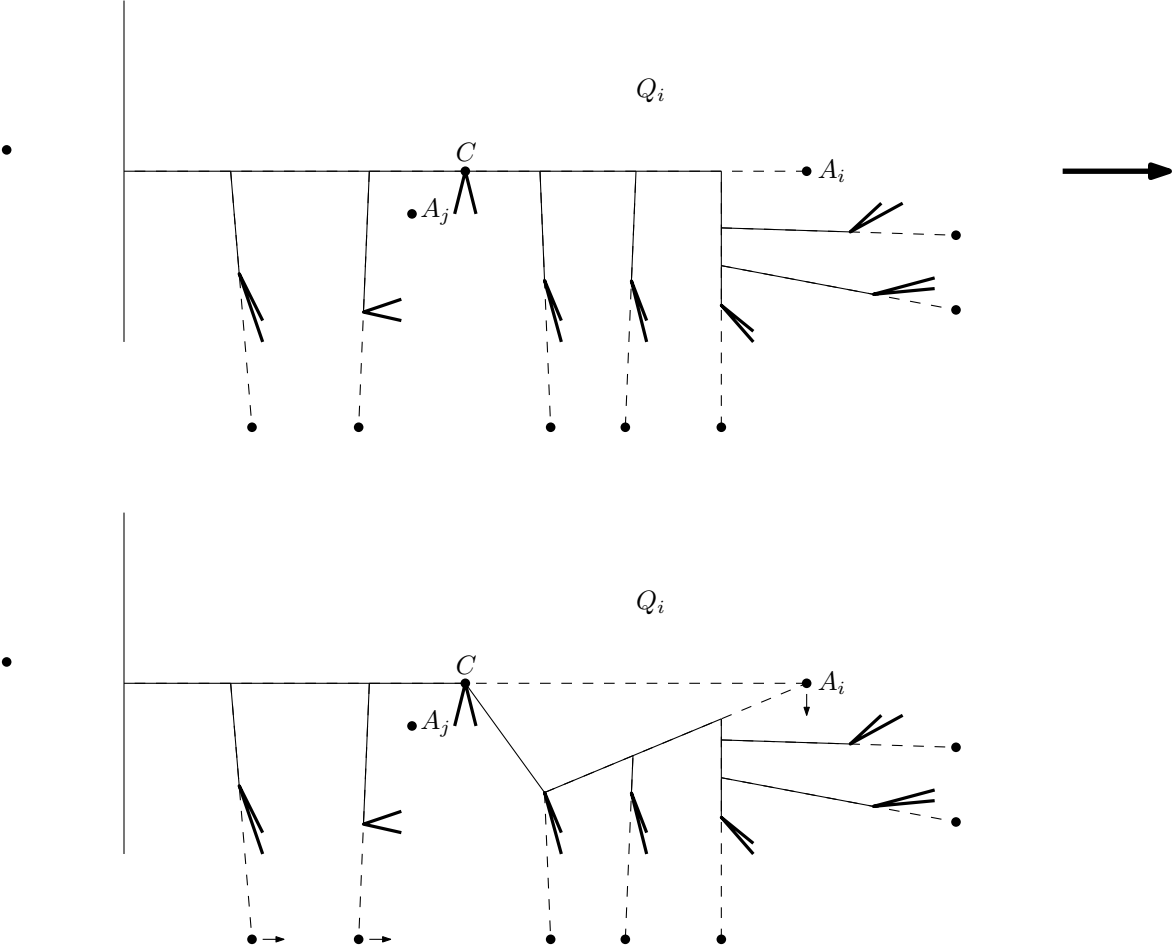

D
F′
F
F″
E
C
Q′3
Q′2
Q′1
G
Qi
Ai
D
F′
F
F″
E
C
Q′3
Q′2
Q′1
G
Qi
Ai
D
F′
F
F″
E
C
Q′3
Q′2
Q′1
G
Qi
Ai

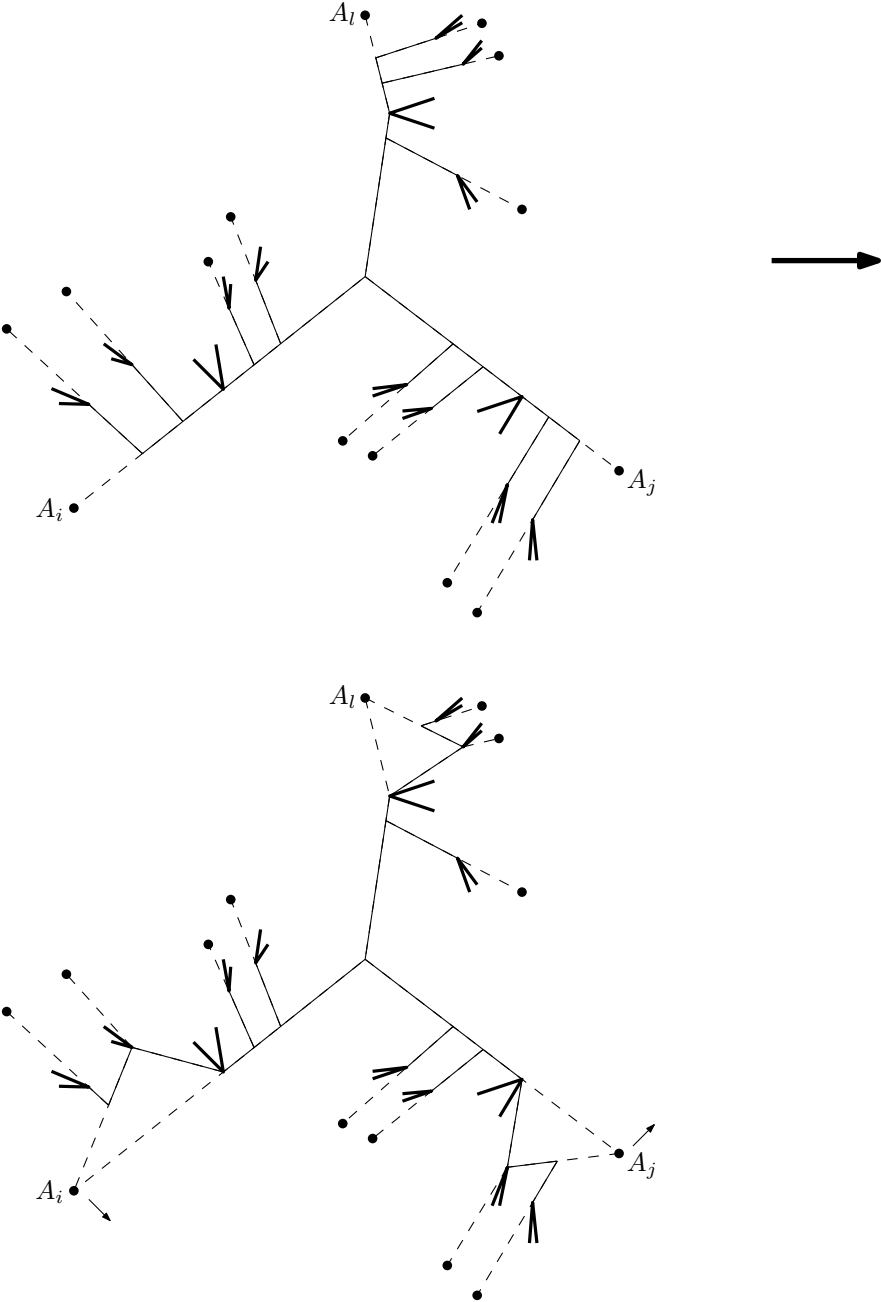

D
F′
F
E
C
Q_p
Q_m
Q_l
Q_i
A_i
D
F′
F
E
C
Q_p
Q_m
Q_l
Q_i
A_i



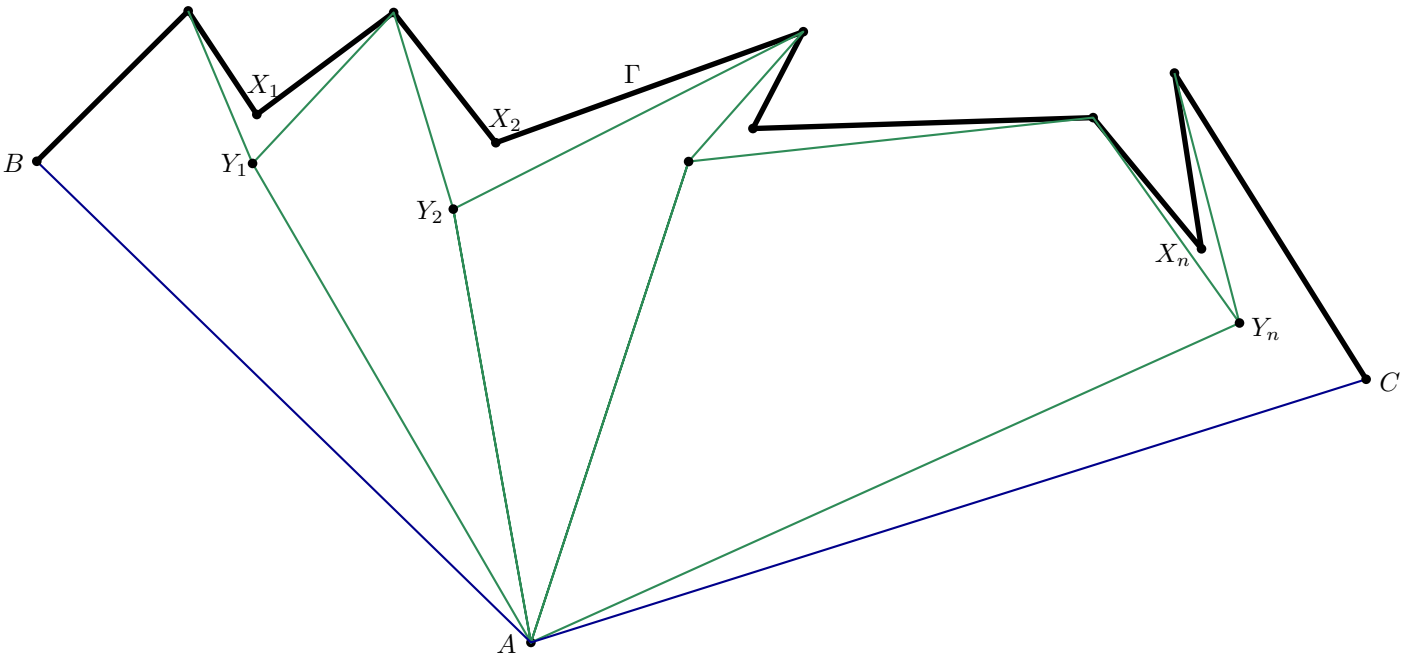

C
D


D
γ
A_i
A_j
C
D
A_i
A_j
C
D
γ
A_i
A_j
C
D
A_i
A_j
C


Q_i
C
A_i
A_j
Q_i
C
A_i
A_j

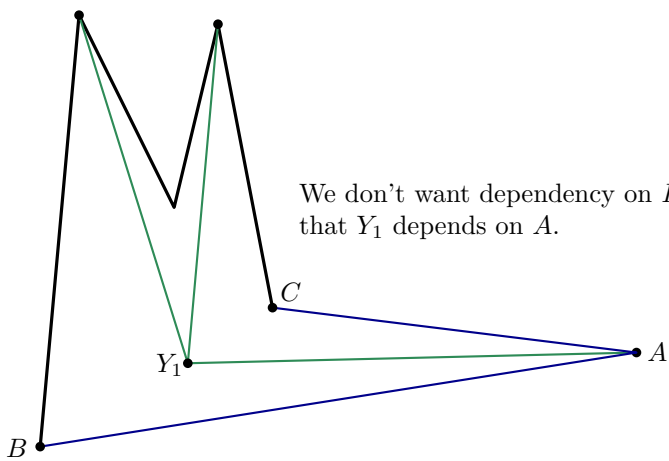

A_l
A_i
A_j
A_l
A_i
A_j

Lemma about Y-shapes:



The thick black curve Γ is fixed. We want to prove that the possible coordinates of the star center A is a convex region defined by lines through the corners of Γ. In other words, A can be *defined* by Γ. We require the green Y-shapes and the blue sightlines to have the combinatorial pattern as seen here, i.e., no intersections.

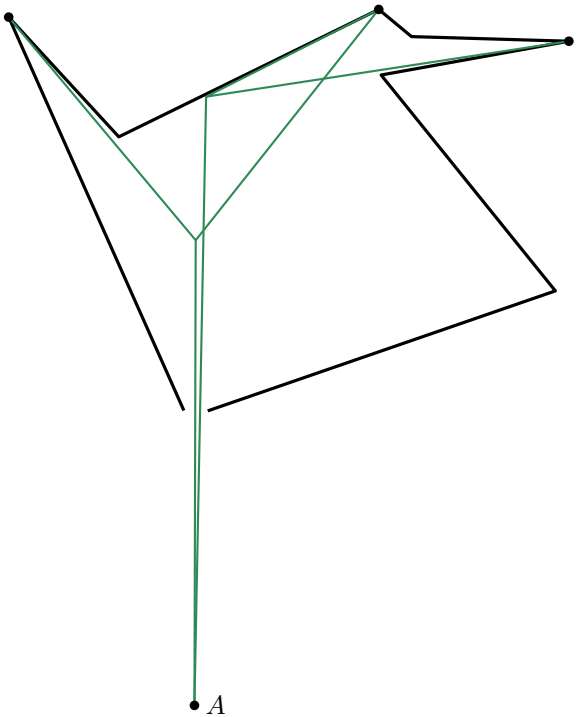
Idea: We first prove this by induction on n: Let us fix A. If one of the endpoints B or C is convex, say B, we can place all the Y-centers $Y_1, \ldots, Y_n$ at nice points defined by the other corners of Γ, avoiding depending on B. Some might also need to depend on A, but A will not depend on the ones that depend on A.

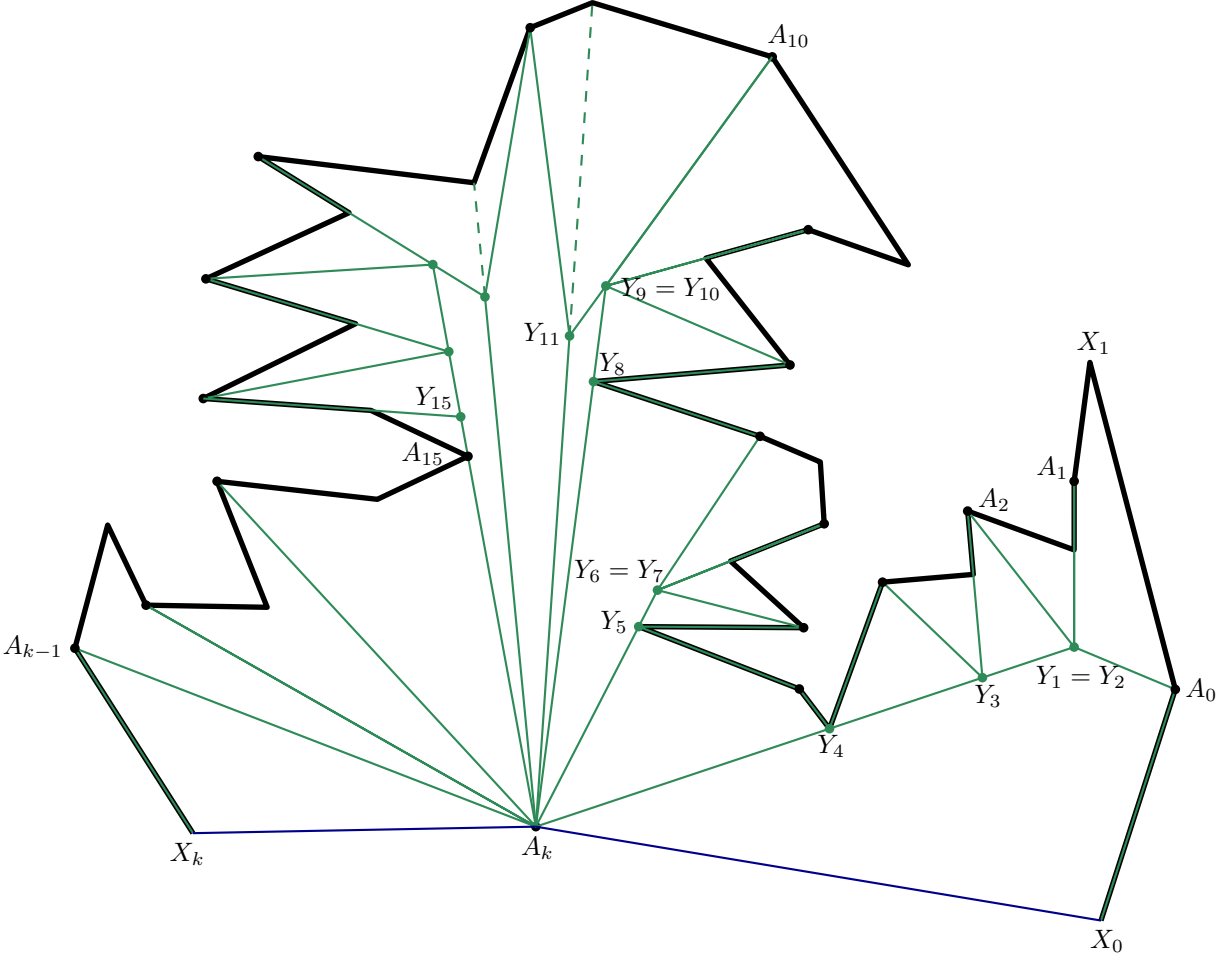


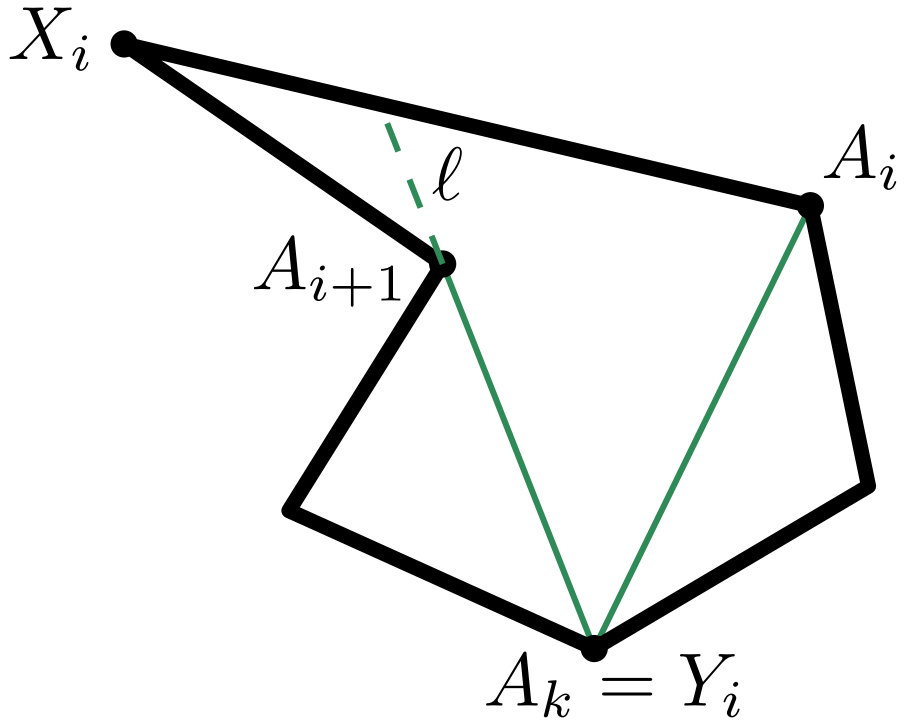
We don't want dependency on B here, since B is convex, so we cannot avoid that Y_1 depends on A.

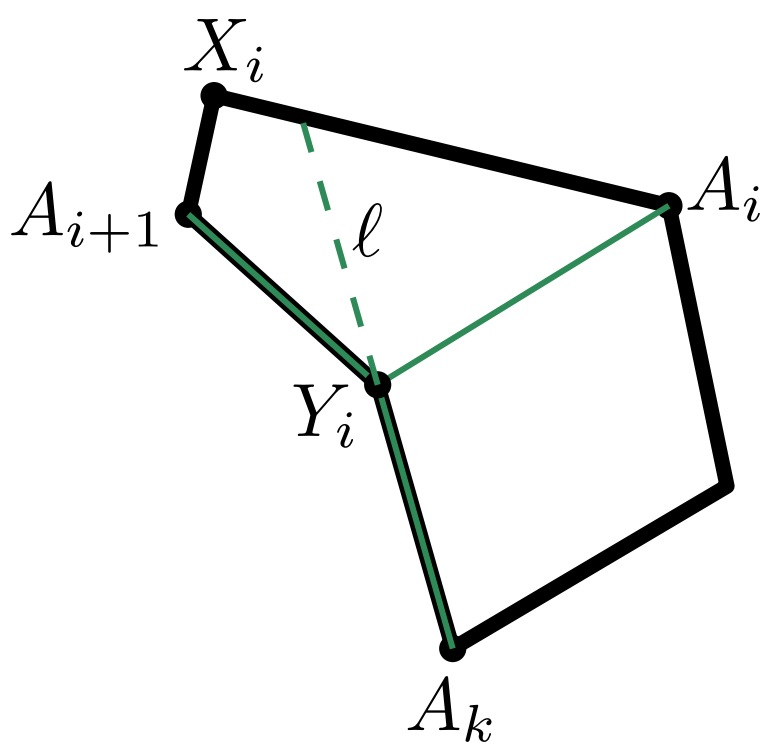
Canonical embedding of Y-shapes: Minimize the distances from Y_i to X_i, or a variant of this (there seem to be no way to define it so that the centers will depend continuously on A and Γ, but it does perhaps not matter).

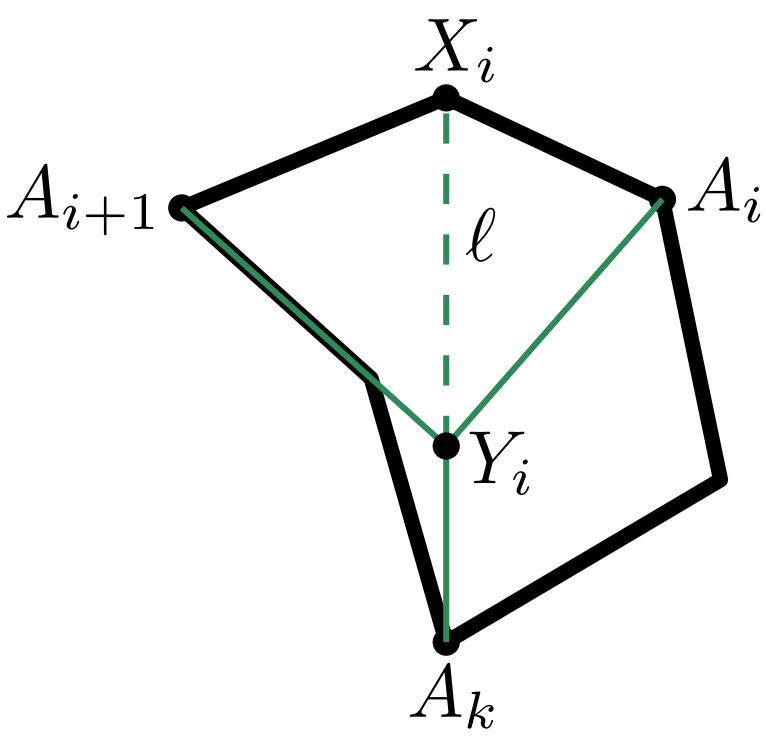
Observation: It is not the case that entangled Y-shapes can always be untangled (for a fixed point A).

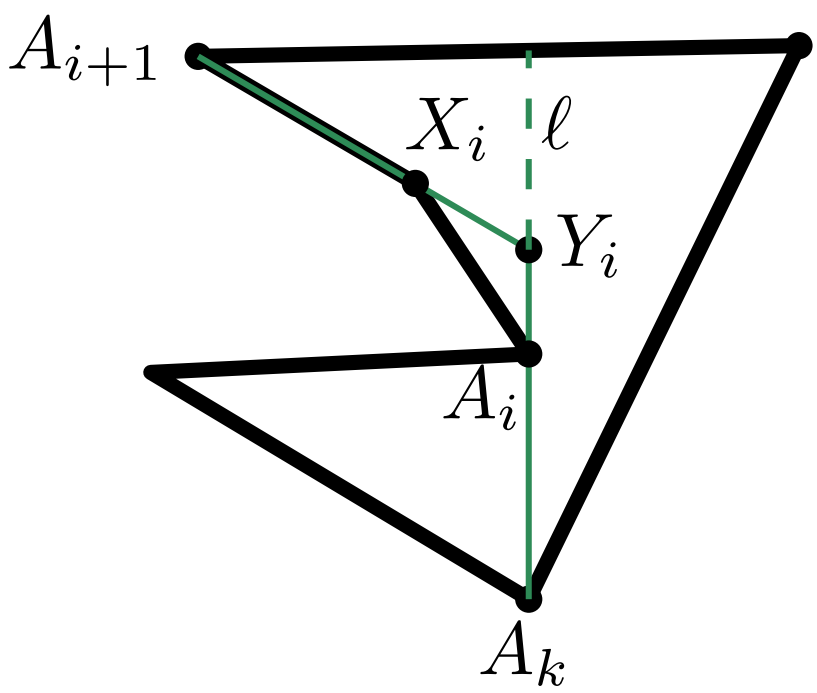


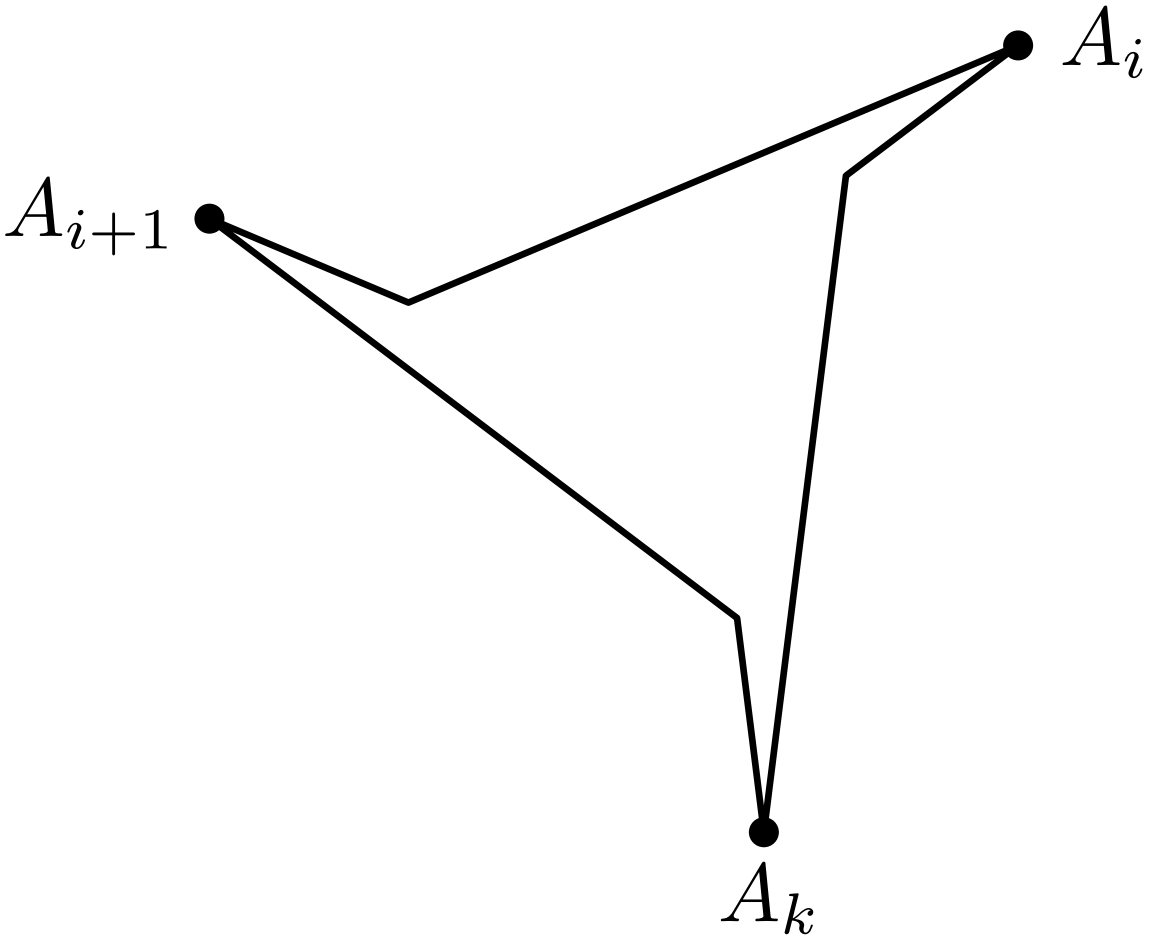

A_{10}
$Y_9 = Y_{10}$
Y_{11}
Y_8
Y_{15}
A_{15}
$Y_6 = Y_7$
Y_5
A_{k-1}
X_k
A_k
Y_4
Y_3
X_1
A_1
A_2
$Y_1 = Y_2$
A_0
X_0

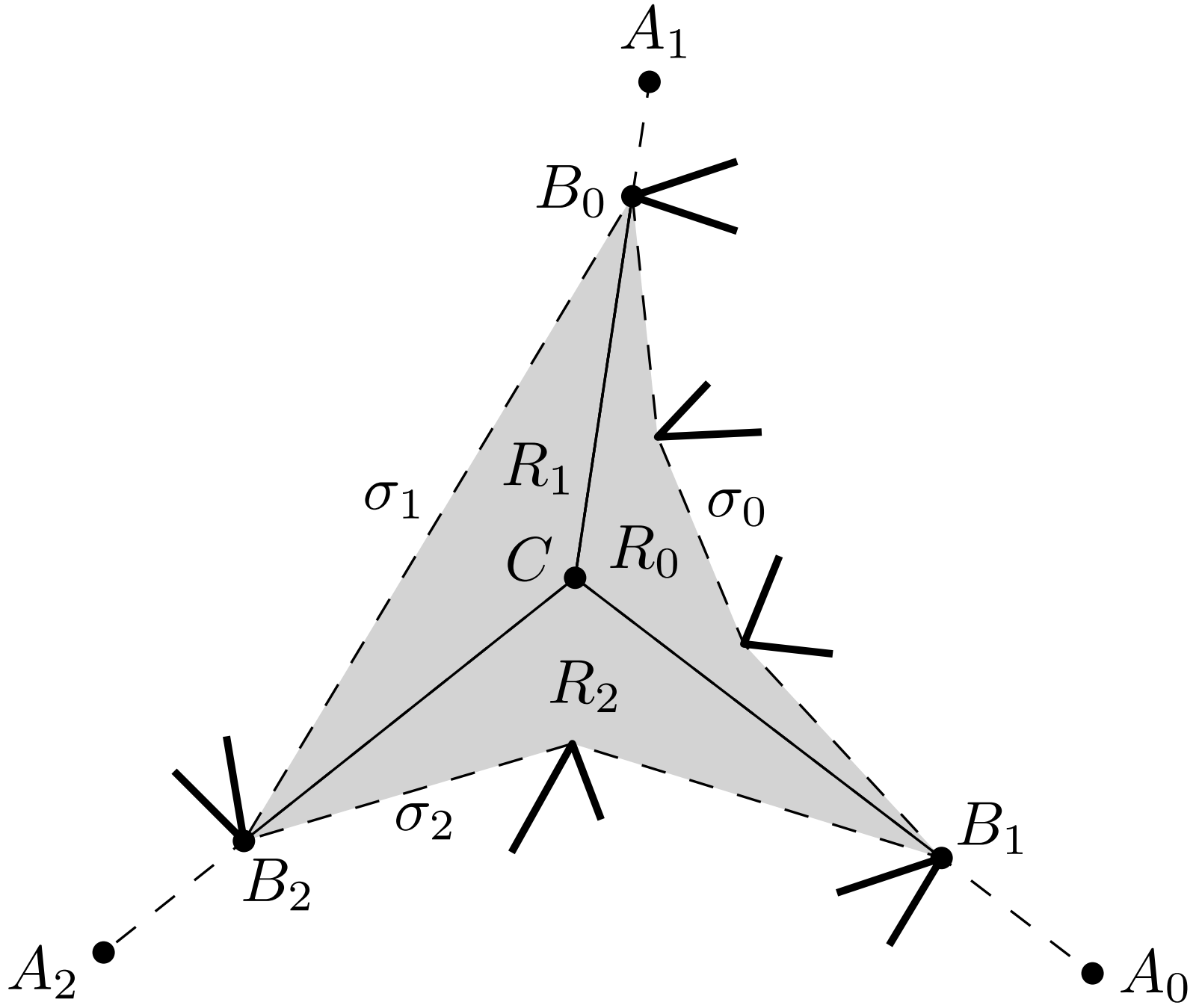

X_i
ℓ
A_i
A_{i+1}
$A_k = Y_i$

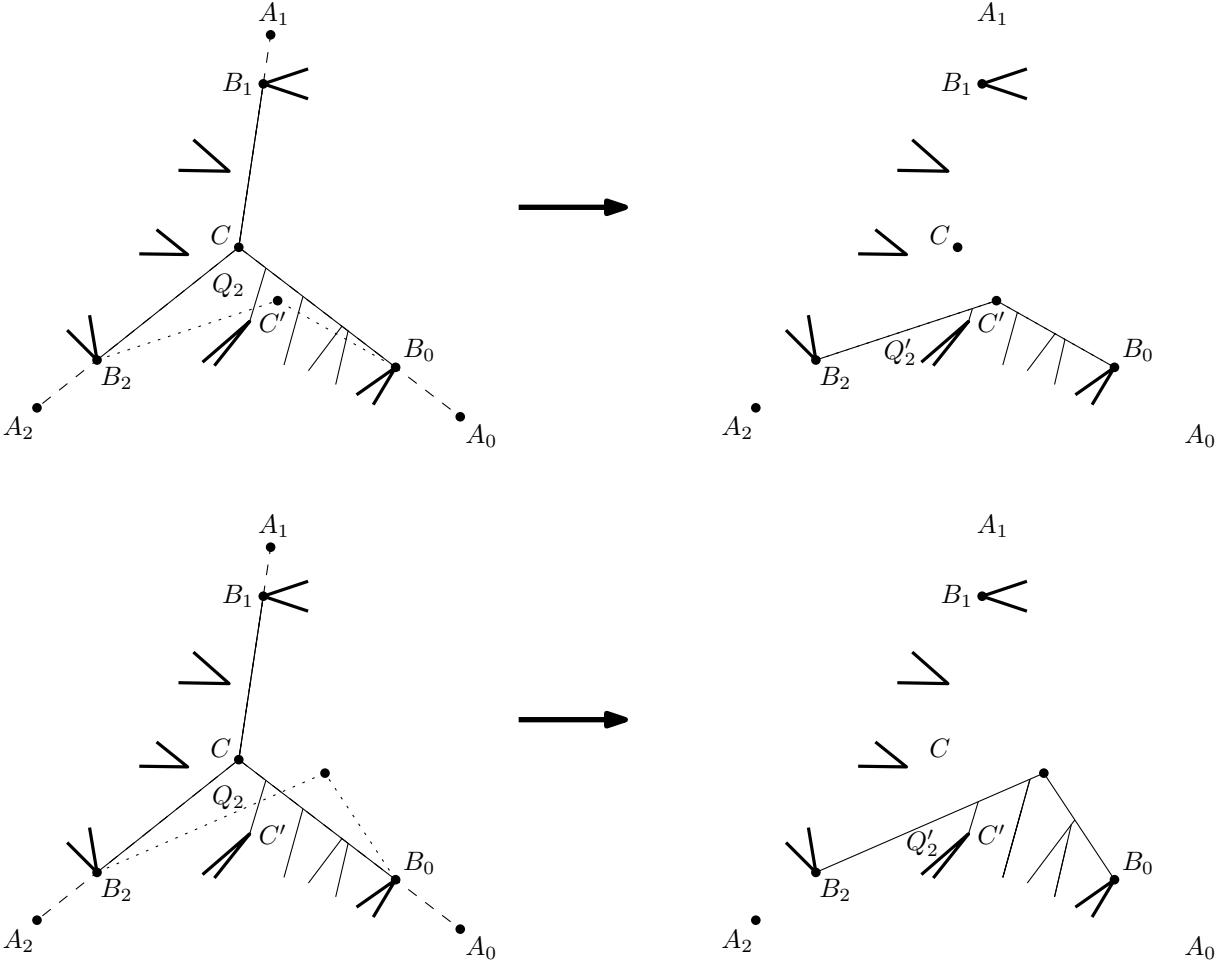

X_i
A_{i+1}
ℓ
A_i
Y_i
A_k

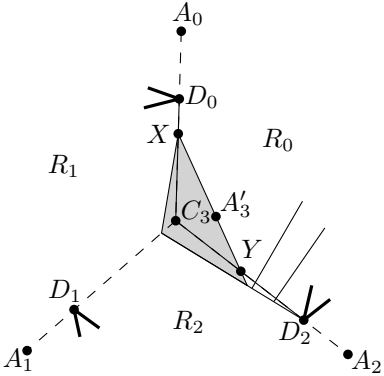

X_i
A_{i+1}
ℓ
A_i
Y_i
A_k

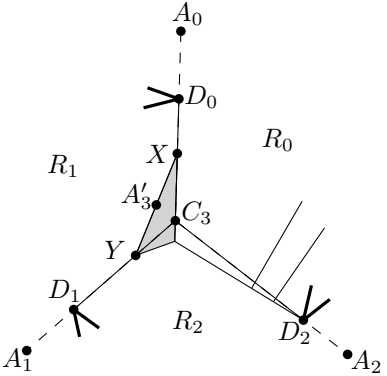

A_{i+1}
X_i
ℓ
Y_i
A_i
A_k

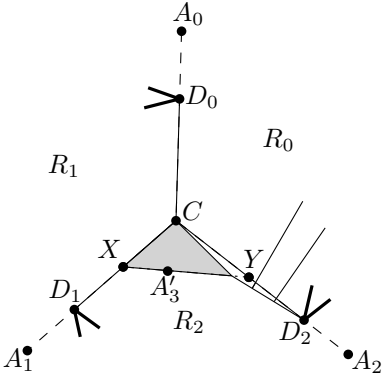

A_i
A_{i+1}
A_k

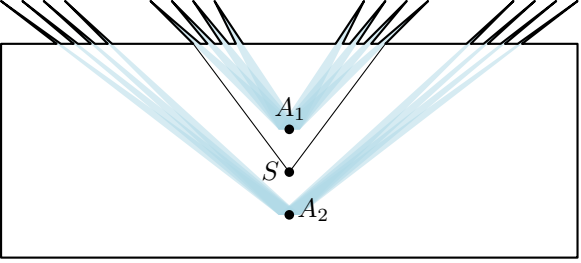

A_1
B_0
R_1
σ_1
σ_0
C
R_0
R_2
σ_2
B_1
B_2
A_2
A_0

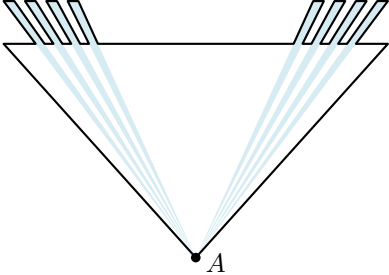

A_1
B_1
C
Q_2
C'
B_0
B_2
A_2
A_0
A_1
B_1
C
C'
Q_2'
B_0
B_2
A_2
A_0
A_1
B_1
C
Q_2
C'
B_0
B_2
A_2
A_0
A_1
B_1
C
Q_2'
C'
B_0
B_2
A_2
A_0

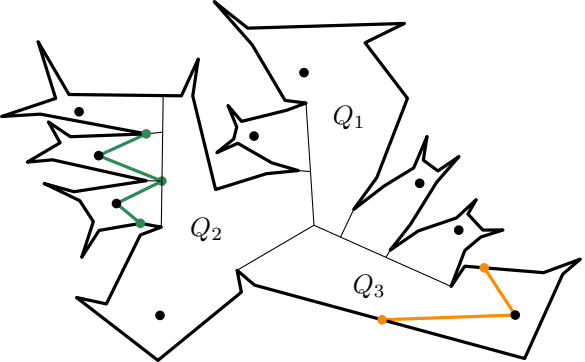

A_0
D_0
X
R_0
R_1
C_3
A'_3
Y
D_1
R_2
D_2
A_1
A_2

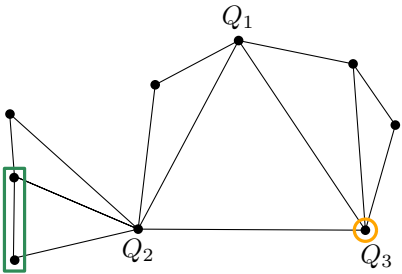

A_0
D_0
R_0
X
R_1
A'_3
C_3
Y
D_1
R_2
D_2
A_1
A_2

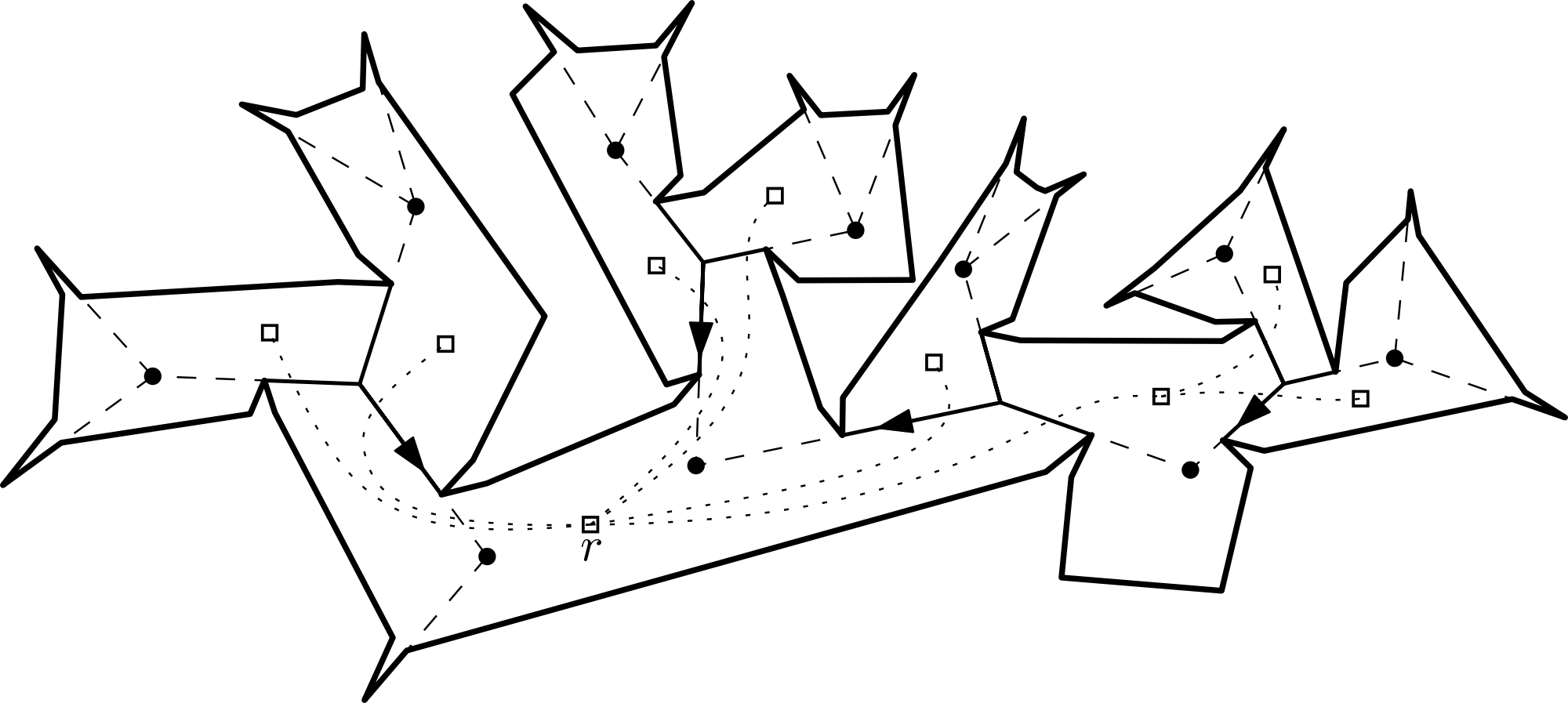

A_0
D_0
R_0
R_1
C
X
Y
D_1
A'_3
R_2
D_2
A_1
A_2

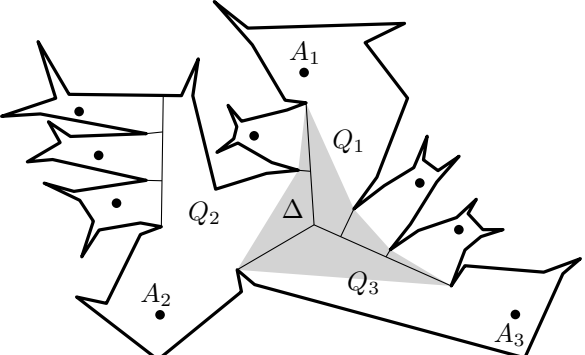

A_1
S
A_2

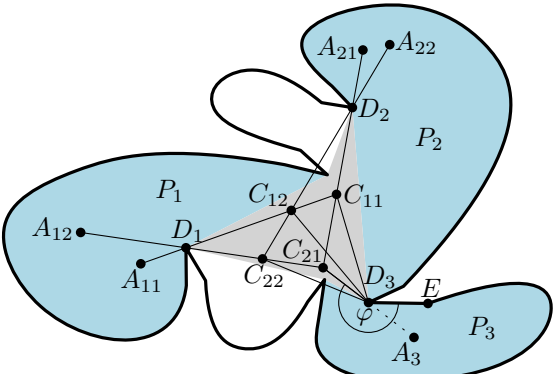

A

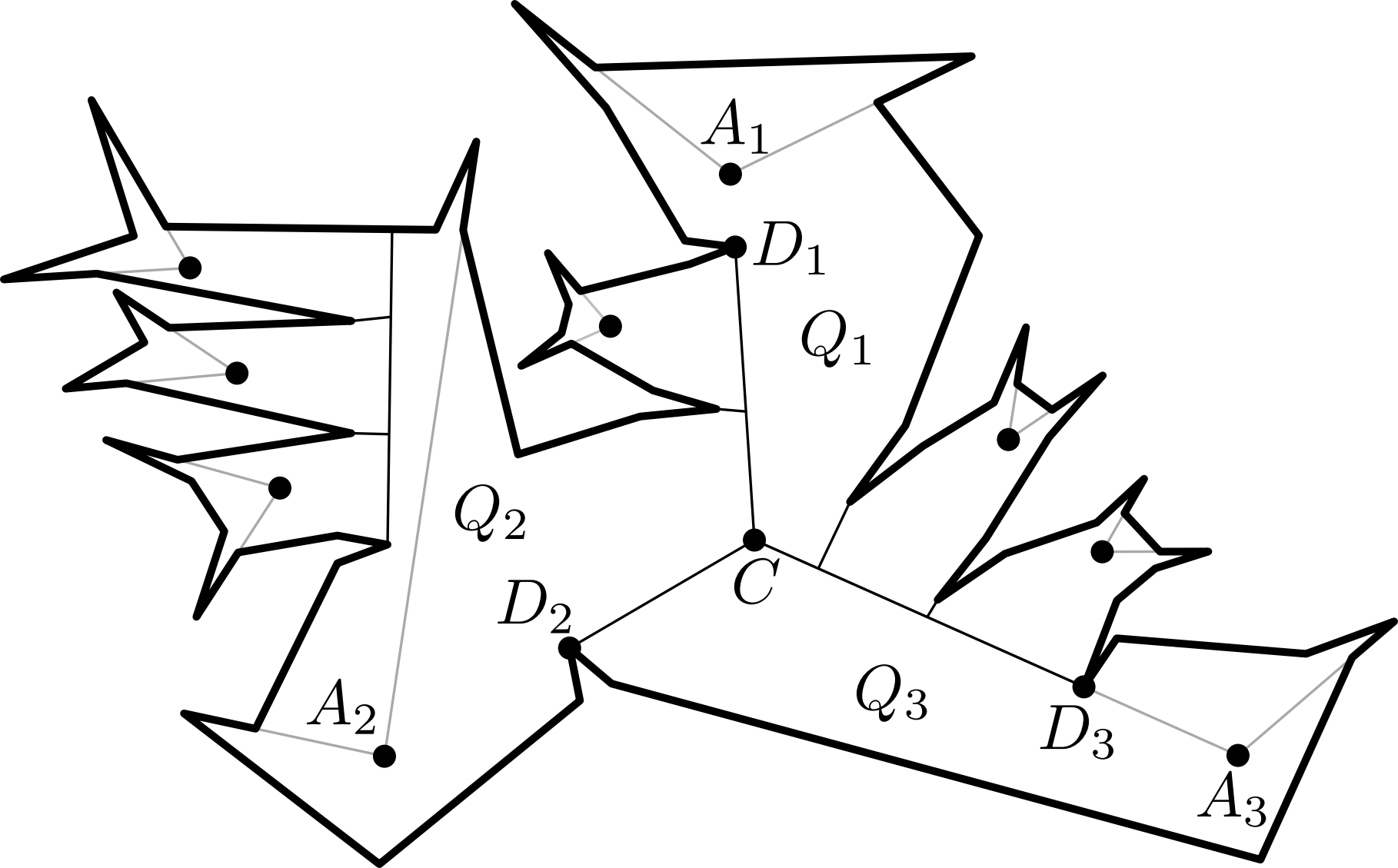

Q_1
Q_2
Q_3

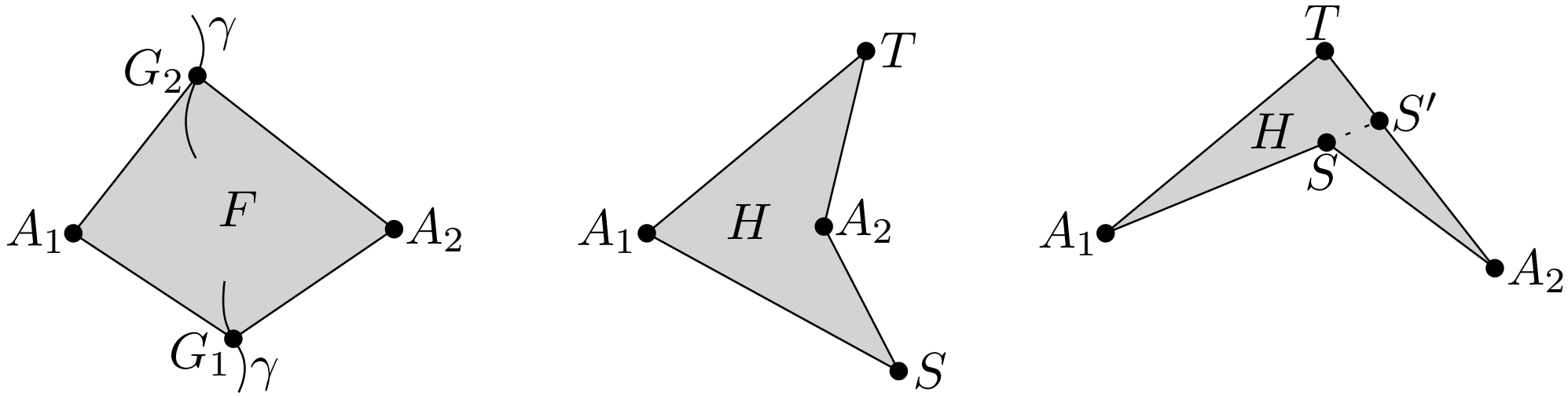

Q_1
Q_2
Q_3

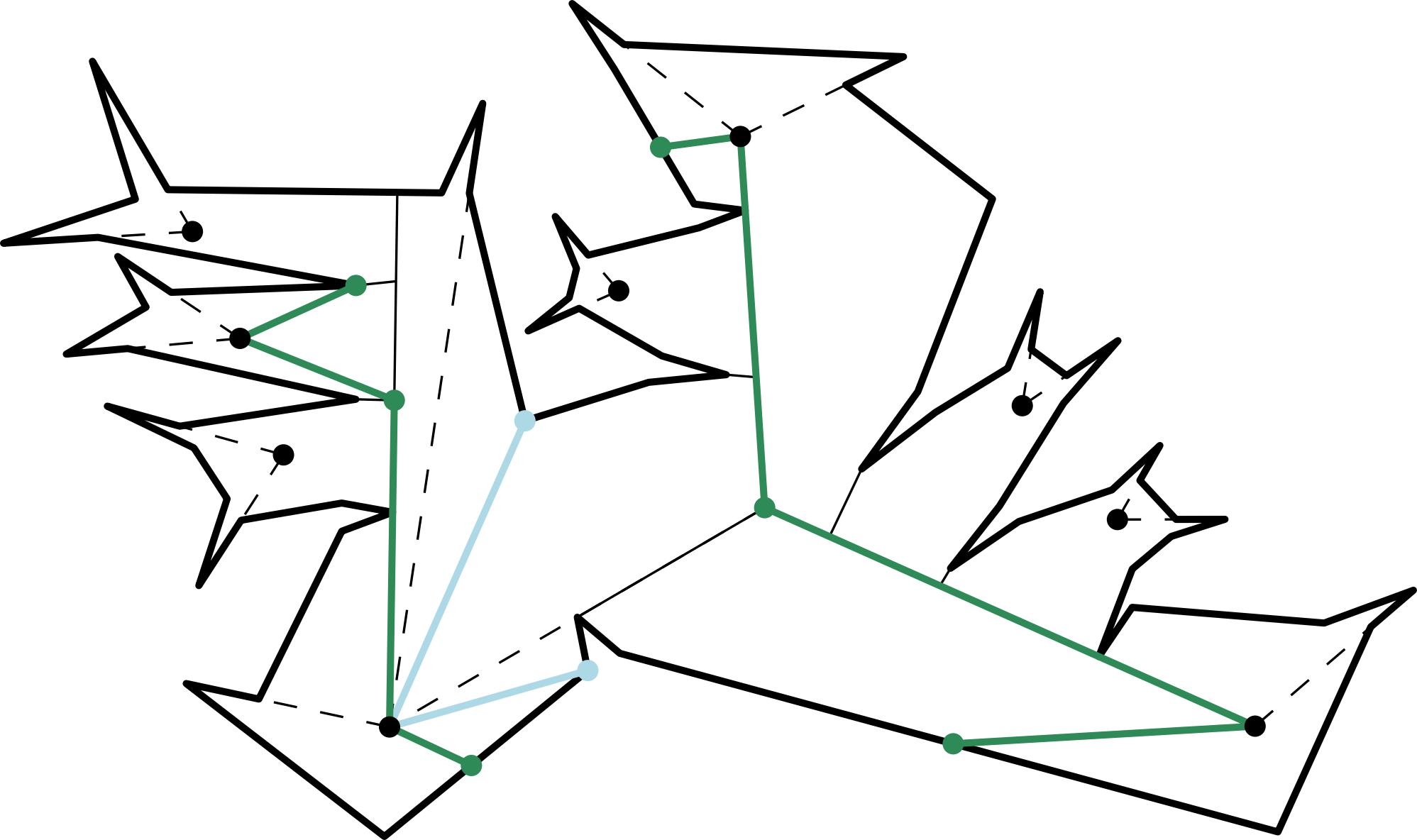

r

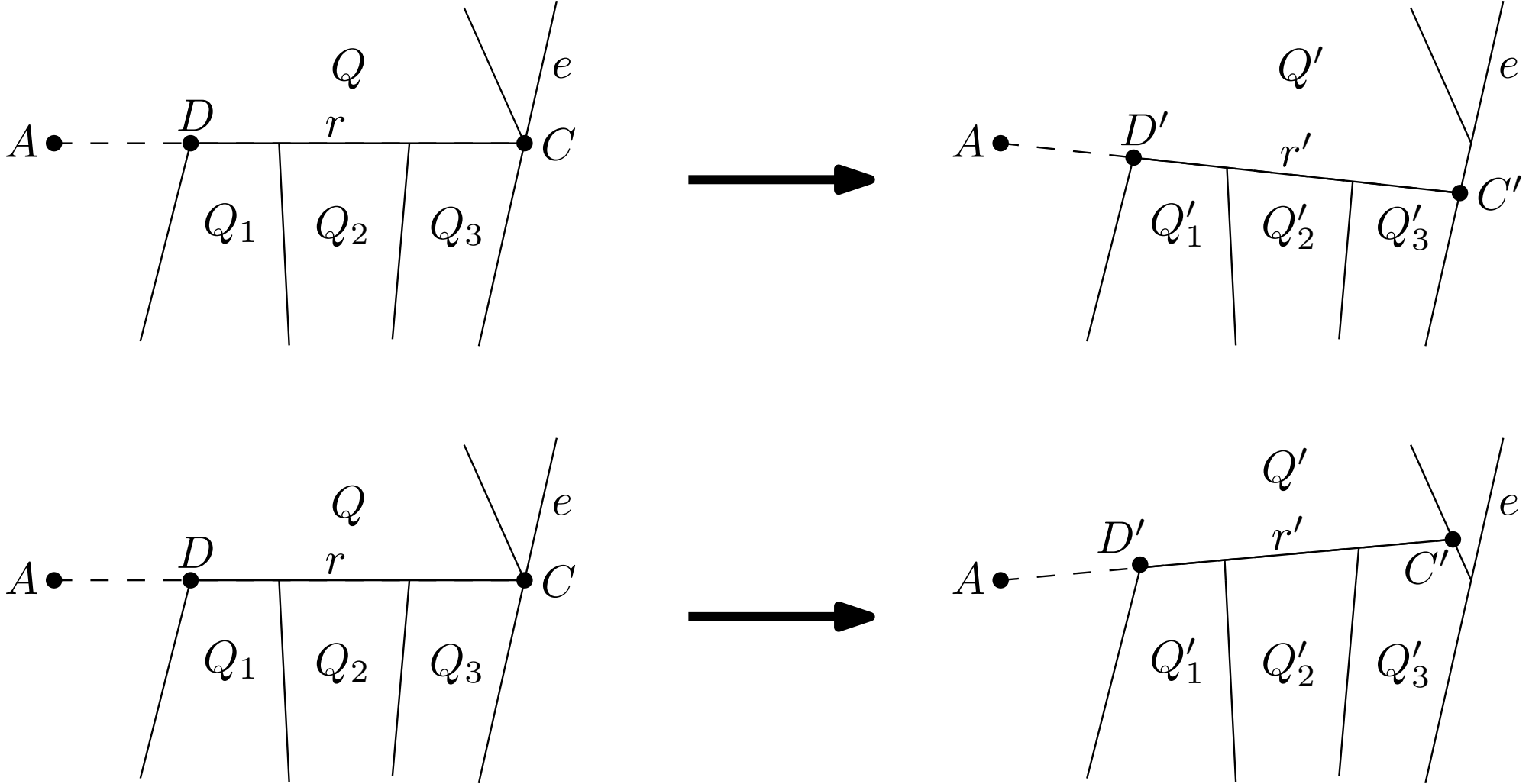

A_1
Q_1
Q_2
Δ
A_2
Q_3
A_3

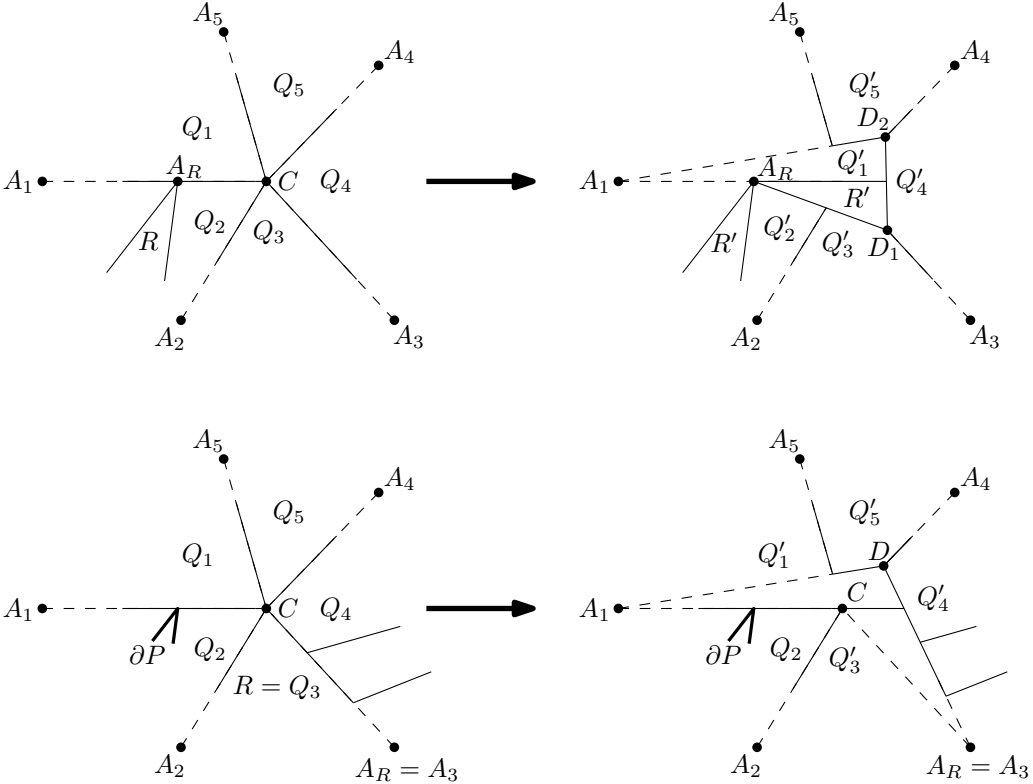

A_{21}
A_{22}
D_2
P_2
P_1
C_{12}
C_{11}
A_{12}
D_1
C_{21}
A_{11}
C_{22}
D_3
E
φ
P_3
A_3

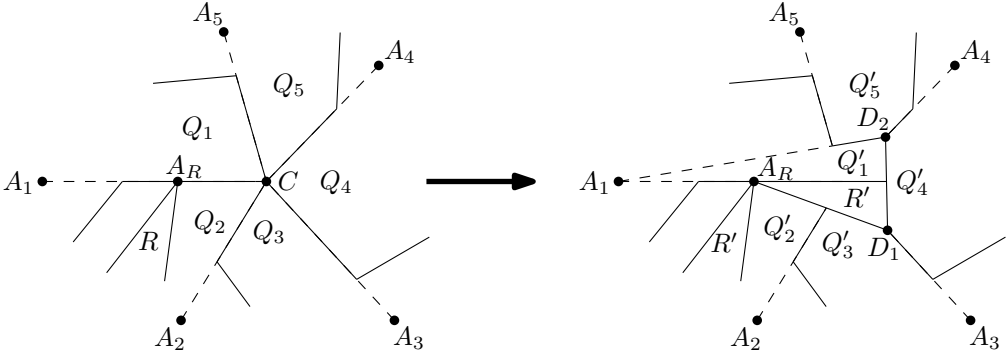

A_1
D_1
Q_1
Q_2
C
D_2
A_2
Q_3
D_3
A_3

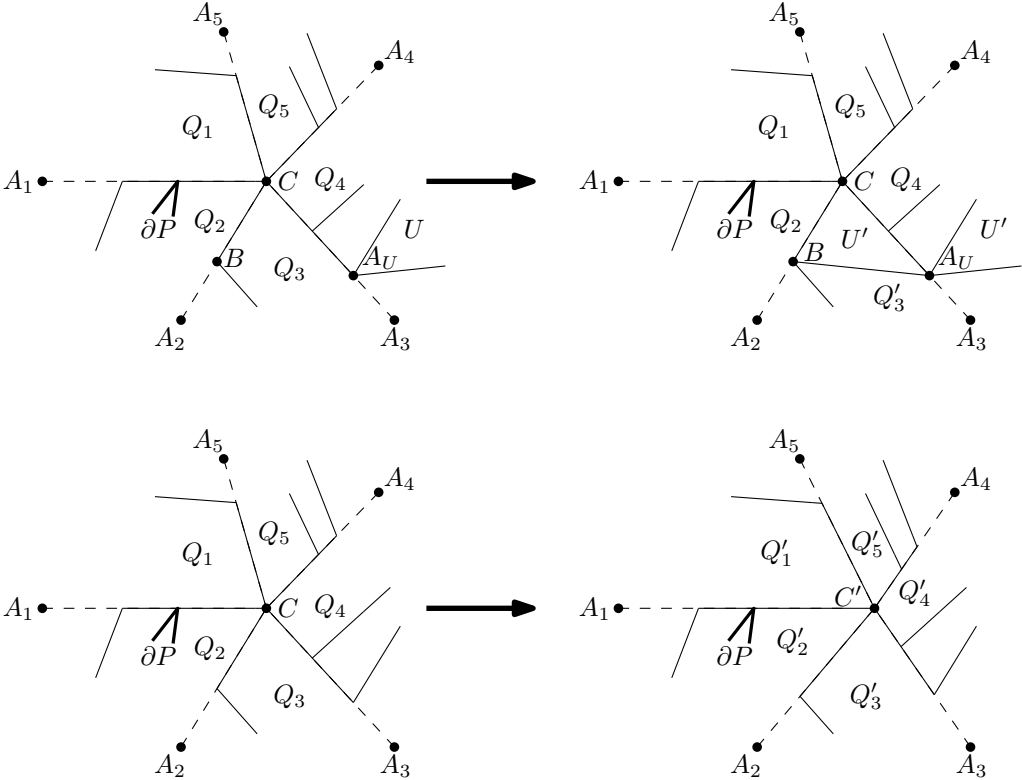

γ
G_2
F
A_1
A_2
G_1
γ
T
H
A_1
A_2
S
T
H
S'
S
A_1
A_2

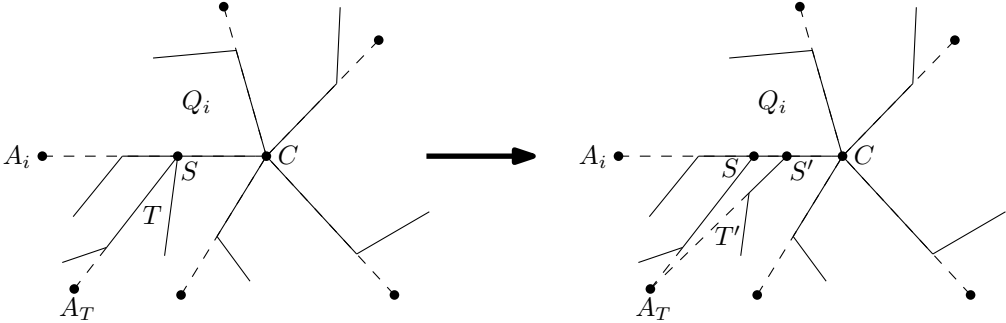


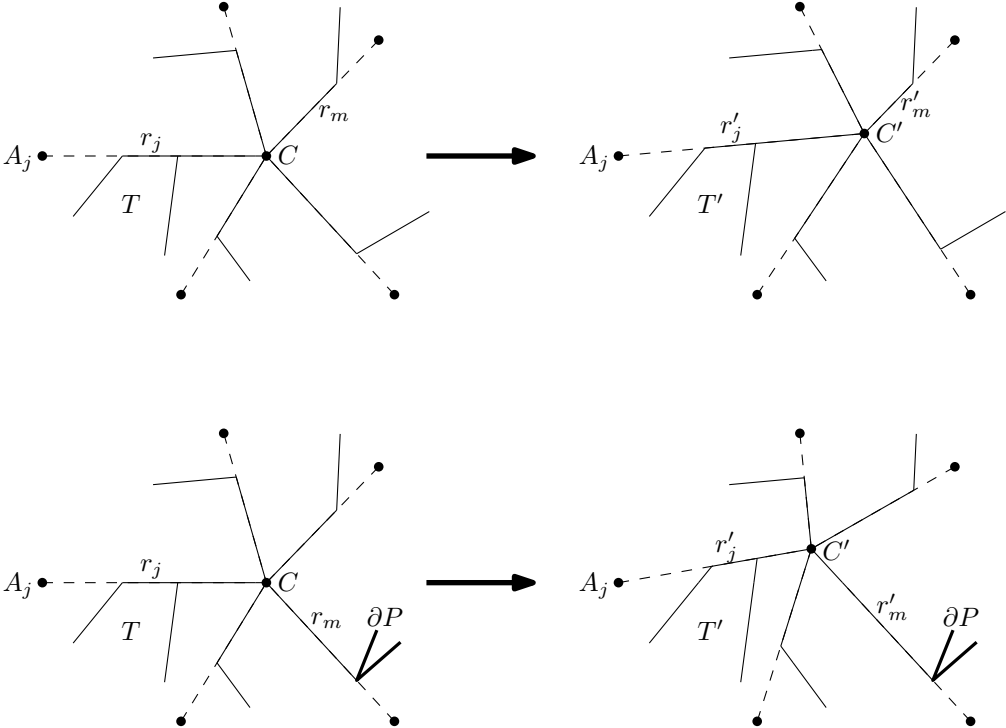

Q
e
A
D
r
C
Q_1
Q_2
Q_3
Q'
e
A
D'
r'
C'
Q'_1
Q'_2
Q'_3
Q
e
A
D
r
C
Q_1
Q_2
Q_3
Q'
e
D'
r'
A
C'
Q'_1
Q'_2
Q'_3

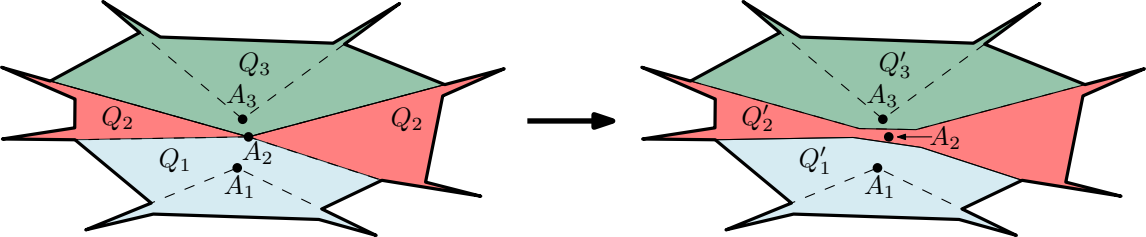

A_5
A_4
Q_5
Q_1
A_1
A_R
C
Q_4
R
Q_2
Q_3
A_2
A_3
A_5
A_4
Q'_5
D_2
A_1
A_R
Q'_1
Q'_4
R'
R'
Q'_2
Q'_3
D_1
A_2
A_3
A_5
A_4
Q_5
Q_1
A_1
C
Q_4
∂P
Q_2
$R = Q_3$
A_2
$A_R = A_3$
A_5
A_4
Q'_5
Q'_1
D
A_1
C
Q'_4
∂P
Q_2
Q'_3
A_2
$A_R = A_3$

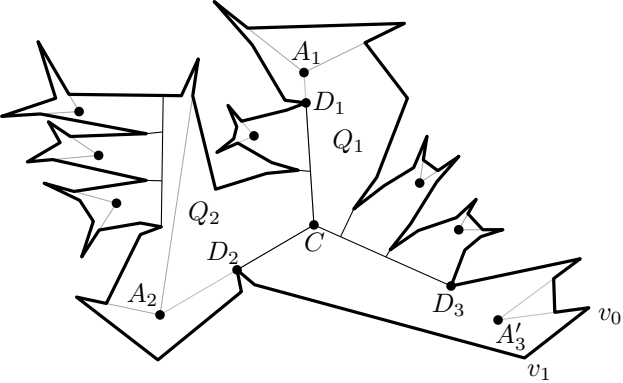

A_5
A_4
Q_5
Q_1
A_1
A_R
C
Q_4
R
Q_2
Q_3
A_2
A_3
A_5
A_4
Q'_5
D_2
A_1
A_R
Q'_1
Q'_4
R'
R'
Q'_2
Q'_3
D_1
A_2
A_3

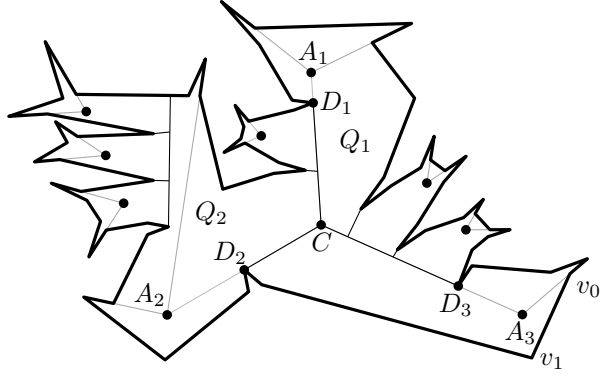

A_5
A_4
Q_1
Q_5
A_1
C
Q_4
∂P
Q_2
U
B
Q_3
A_U
A_2
A_3
A_5
A_4
Q_1
Q_5
A_1
C
Q_4
∂P
Q_2
U'
U'
B
A_U
Q_3'
A_2
A_3
A_5
A_4
Q_1
Q_5
A_1
C
Q_4
∂P
Q_2
Q_3
A_2
A_3
A_5
A_4
Q_1'
Q_5'
C'
Q_4'
A_1
∂P
Q_2'
Q_3'
A_2
A_3

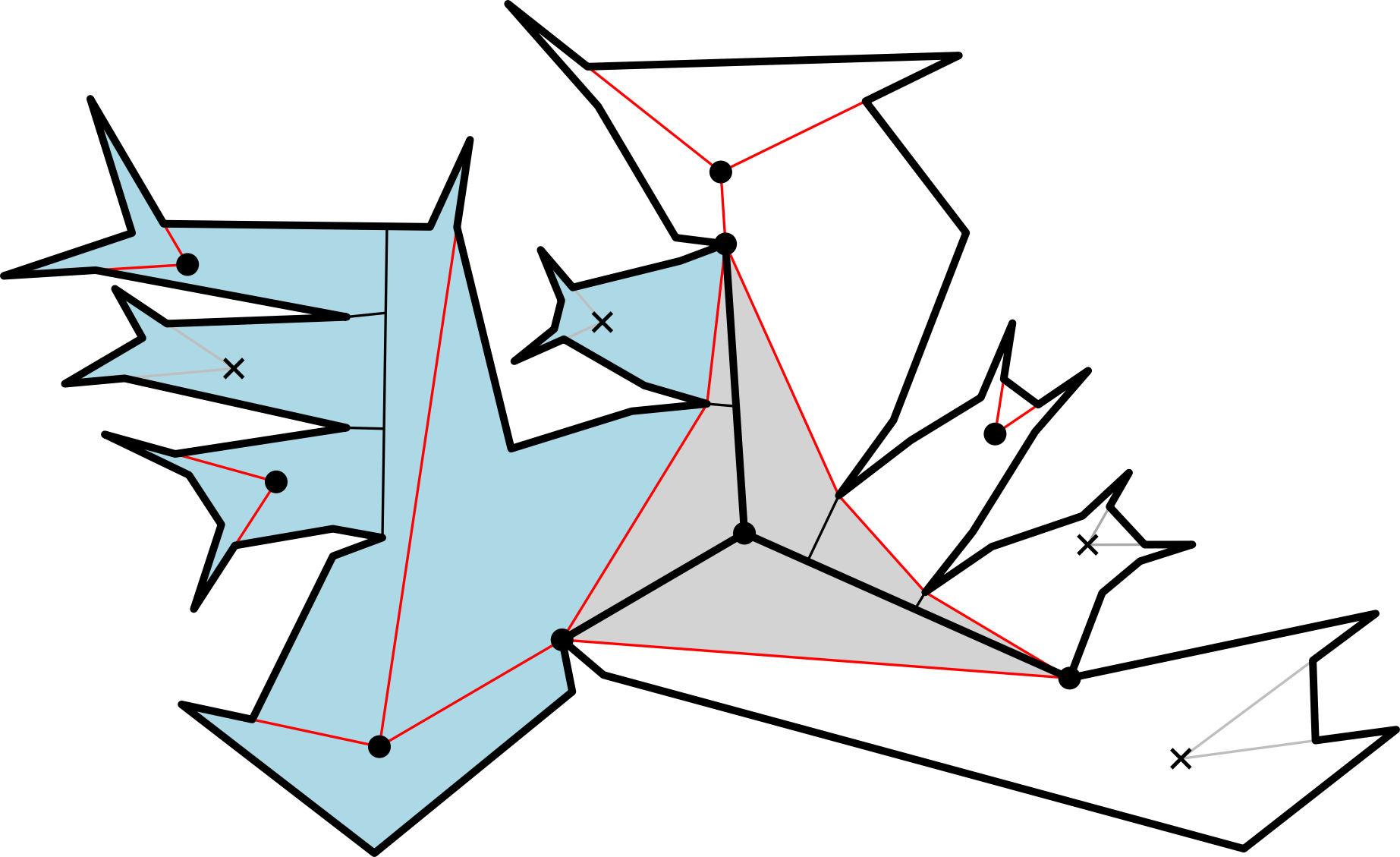

Q_i
A_i
S
C
T
A_T
Q_i
A_i
S
S'
C
T'
A_T

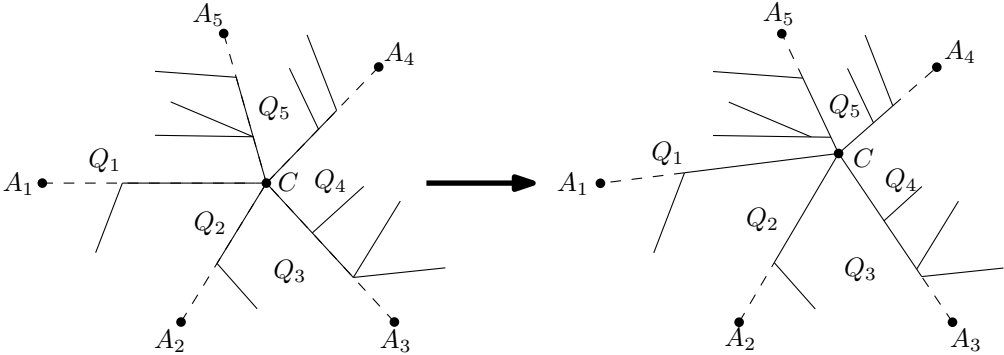

A_j
r_j
C
r_m
T
A_j
r'_j
C'
r'_m
T'
A_j
r_j
C
r_m
∂P
T
A_j
r'_j
C'
r'_m
∂P
T'

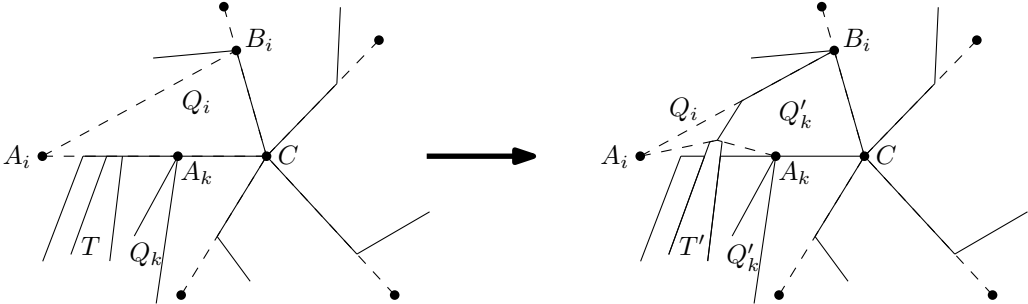

Q_3
A_3
Q_2
Q_2
Q_1
A_2
A_1
Q'_3
A_3
Q'_2
A_2
Q'_1
A_1

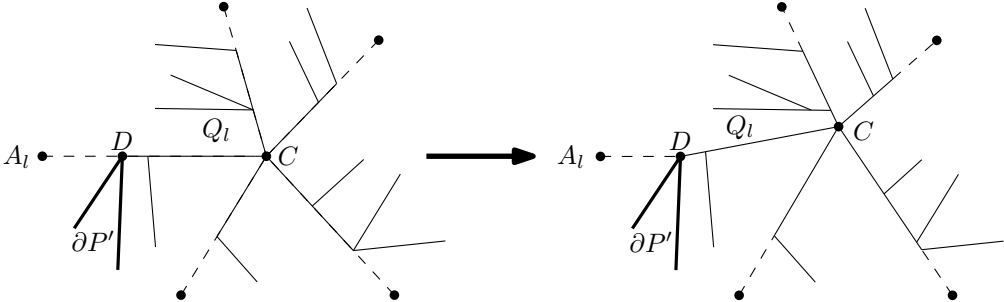

A_1
D_1
Q_1
Q_2
C
D_2
A_2
D_3
A'_3
v_0
v_1

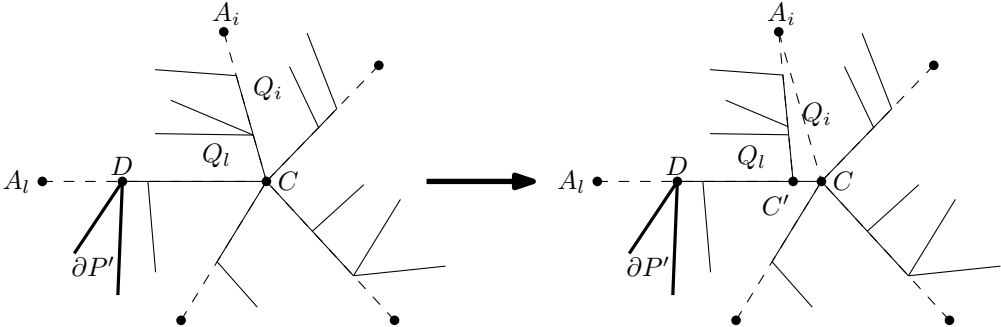

A_1
D_1
Q_1
Q_2
C
D_2
A_2
D_3
A_3
v_0
v_1

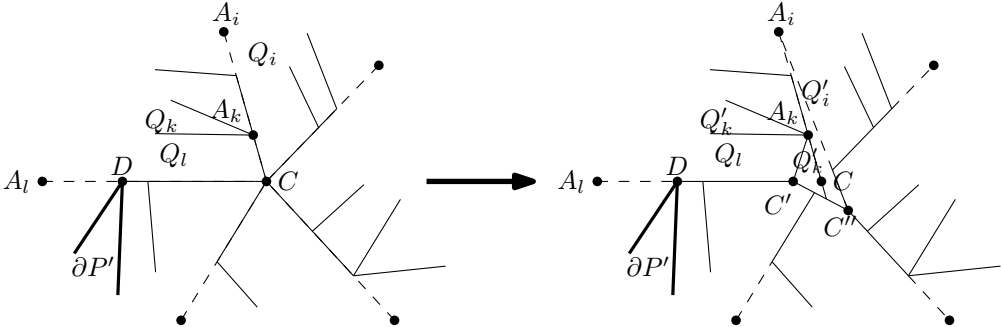


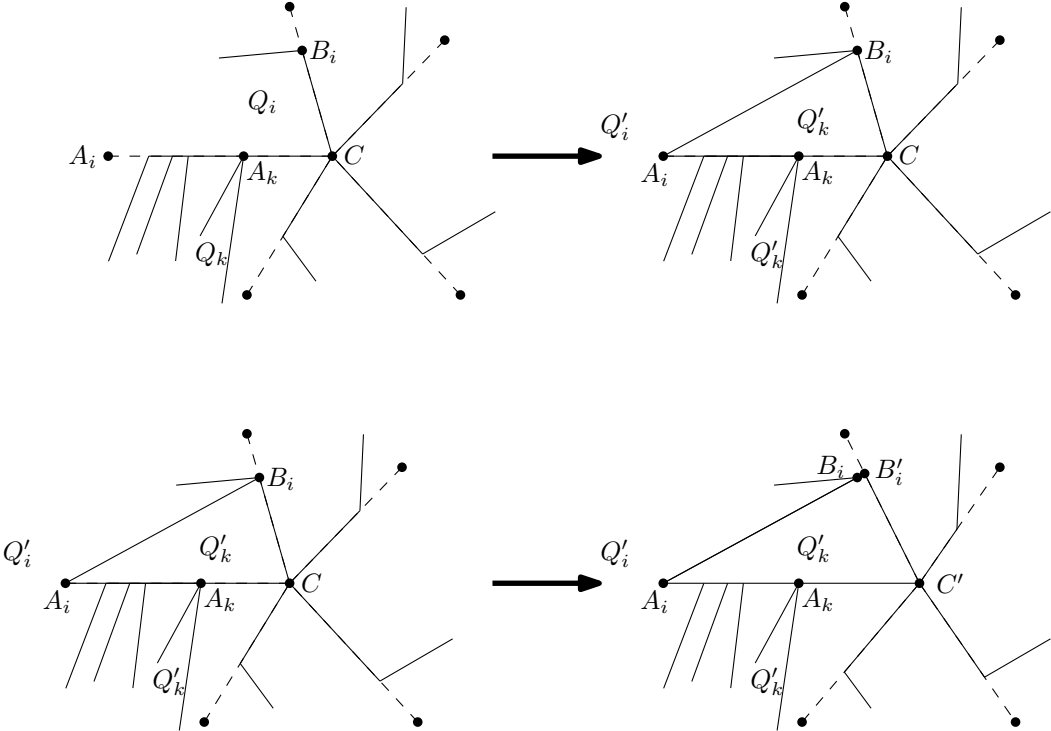

A_5
A_4
Q_5
Q_1
A_1
C
Q_4
Q_2
Q_3
A_2
A_3
A_5
A_4
Q_5
Q_1
A_1
C
Q_4
Q_2
Q_3
A_2
A_3


B_i
Q_i
A_i
A_k
C
T
Q_k
B_i
Q_i
Q'_k
A_i
A_k
C
T'
Q'_k


A_l
D
Q_l
C
$\partial P'$
A_l
D
Q_l
C
$\partial P'$


A_i
Q_i
Q_l
A_l
D
C
$\partial P'$
A_i
Q_i
Q_l
A_l
D
C
C'
$\partial P'$


A_i
Q_i
A_k
Q_k
Q_l
A_l
D
C
$\partial P'$
A_i
Q'_i
A_k
Q'_k
Q_l
Q'_l
C
A_l
D
C'
C''
$\partial P'$


B_i
Q_i
A_i
A_k
C
Q_k
Q'_i
B_i
Q'_k
A_i
A_k
C
Q'_k
Q'_i
B_i
Q'_k
A_i
A_k
C
Q'_k
Q'_i
B_i
B'_i
Q'_k
A_i
A_k
C'
Q'_k